

Exercise-Based and Nutrition-Based Prehabilitation Programs in Surgery: A Systematic Review and Meta-Analysis

Catherine T Cascavita, MD, MLS¹, Anne E Hall, BS¹, Kaavian Shariati, MEng, BS¹, Jose M Chevalier, BS¹, Alexander A Argame, BSN, RN^{1,2}, Nghiem H Nguyen, BA^{1,2}, Chi-Hong Tseng, PhD³, Marco A Hidalgo, PhD^{3,4}, Justine C Lee, MD, PhD, FACS^{1,4}, and the Craniofacial Outcomes Research Team¹

¹Division of Plastic and Reconstructive Surgery, Department of Surgery, University of California Los Angeles, David Geffen School of Medicine and the Greater Los Angeles VA Healthcare System, Los Angeles, CA, USA

²Kaiser Permanente Bernard J. Tyson School of Medicine, Pasadena, CA, USA

³Division of General Internal Medicine & Health Services Research, Internal Medicine-Pediatrics and Preventive Medicine Section, University of California Los Angeles, David Geffen School of Medicine, Los Angeles, CA USA

⁴UCLA Gender Health Program, University of California Los Angeles, David Geffen School of Medicine, Los Angeles, CA, USA

Disclosure Information: Nothing to disclose.

Disclosures outside the scope of this work: Dr Lee is a paid consultant for Stryker. Other authors have nothing to disclose.

Support: This work was supported by the Bernard G Sarnat Endowment for Craniofacial Biology (JCL), the Jean Perkins Foundation (JCL), and the National Center for Advancing Translational Science (NCATS) of the National Institutes of Health under the UCLA Clinical and Translational Science Institute grant [UL1TR001881]. All sources of funds supporting the

completion of this manuscript are under the auspices of the University of California, Los Angeles.

Disclaimer: The funders had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Corresponding Author: Justine C. Lee, MD, PhD, FACS University of California, Los Angeles
Division of Plastic and Reconstructive Surgery 200 Medical Plaza, Suite 460 Los Angeles, CA
90095-6960 Telephone: (310) 794-7616 Fax: (310) 206-6833 justine@ucla.edu

Short Running Head: Prehabilitation in Surgery

Background: Exercise and nutrition-based prehabilitation programs are promising emerging strategies for enhancing surgical recovery. However, prior studies have reported mixed findings regarding their impact on postoperative outcomes. The purpose of this study is to investigate the efficacy of exercise and nutrition-based prehabilitation on postoperative outcomes, including length-of-stay (LOS), complications, quality-of-life (QoL), pain, and mental health.

Methods: A systematic review and meta-analysis was conducted, using MEDLINE, Cochrane Central, EMBASE, and Web of Science, following PRISMA guidelines. Studies published between 2004 and 2024 were identified with search terms including “prehabilitation,” “exercise,” “nutrition,” and “surgery.” Eligible studies (n=293) were randomized controlled trials (RCTs) evaluating exercise or nutrition-based prehabilitation programs and reporting LOS and complications as primary outcomes. Secondary outcomes included QoL, pain, and mental health scores. Random-effects meta-analyses estimated pooled effect sizes.

Results: 23 RCTs were included (2,182 participants). Exercise or nutrition-based prehabilitation reduced complications (odds ratio (OR) 0.52, 95% confidence interval (CI):0.35,0.78, $p<0.002$, $I^2=47.1\%$) and LOS (mean difference (MD) -0.44 days, 95% CI:-0.78,-0.11, $p=0.01$, $I^2=44.8\%$) compared to standard treatment. When comparing interventions, nutrition only had a greater reduction in LOS than exercise only (MD -1.09 days, 95% CI:-1.72,-0.47 vs. MD -0.20 days, 95% CI: -0.51, 0.09; $p=0.01$). While QoL was not reported in nutrition only RCTs, exercise alone improved QoL measures (standardized mean difference 0.94, 95% CI: 0.24,1.64, $p=0.01$, $I^2=93.6\%$) compared to standard treatment.

Conclusion: Exercise or nutrition-based prehabilitation programs reduce LOS and complications across multiple surgical contexts. Further work is needed to elucidate the impact of different intervention protocols and their impact on postoperative outcomes.

Keywords: prehabilitation; surgery; exercise; nutrition

Introduction

Over 300 million surgeries are performed annually worldwide and are expected to increase as our aging population grows.^{1,2} Alarming, up to 25% of surgical patients may experience postoperative complications, contributing to lengthened hospital stays and increased healthcare burdens.³ These poor surgical outcomes are suggested to be influenced by preoperative status and modifiable risk factors, including lessened physical fitness, smoking, uncontrolled diabetes, malnutrition, and poor mental health.⁴⁻⁶ Prehabilitation is an emerging strategy to promote preoperative optimization and mitigate these risks.^{4,5,7-9} Prehabilitation aims to optimize patients preoperatively by improving their physical endurance, nutritional reserves, and mental readiness. Most often multimodal in design, these programs typically encompass exercise, nutrition, psychological intervention, and education components.¹⁰

Among such modalities, exercise is commonly employed and is intended to improve physical fitness, flexibility, and muscle mass.¹¹ Whitney et al. previously found that total hip arthroplasty patients who were more physically active before surgery regained mobility more rapidly after surgery, experiencing shorter hospital stays.¹² Similarly, in a small two-arm cohort study, Nielson et al. found that a two-month exercise-based prehabilitation program coupled with early rehabilitation after spine surgery shortened hospital length of stay (LOS).¹³ Beyond exercise, prehabilitation also frequently features nutrition-focused interventions, as perioperative malnutrition is associated with higher complication rates, LOS, and morbidity.¹⁴ Recognizing the importance of preoperative nutrition optimization, Enhanced Recovery After Surgery (ERAS) guidelines recommend nutritional supplementation at least seven days before surgery in at-risk patients.¹⁵ Generally, perioperative nutritional goals include meeting energy and protein requirements of 25-30 kcal/kg and 1.5g/kg of ideal body weight.¹⁶

Despite growing evidence supporting the value of prehabilitation, the collective effects reported by previous reviews on clinical outcomes have varied.¹⁷ Prior systematic reviews and meta-analyses assessing the impact of prehabilitation on surgical outcomes have encountered conflicting data and have reported inconclusive findings regarding LOS, postoperative complications, and patient-reported anxiety and depression.^{18,19} Additionally, the impact of preoperative nutritional interventions on surgical outcomes was studied in 2024 by Sowerbutts et al., finding that trials evaluating enteral and oral nutrition were inconclusive given low-certainty evidence.^{20,21} While comprehensive, this review was limited to randomized controlled trials (RCTs) of gastrointestinal surgery patients often presenting with malignant pathology, constraining generalizability of the results to a broader, diverse population presenting with primarily surgery-related needs. Beyond the independent benefits of exercise or nutrition-based programs, their combined effects have received less study and are of even greater uncertainty.

Assessing the impact of nutrition and exercise prehabilitation on surgical patients may clarify the benefits of these programs. This systematic review and meta-analysis examine the effects of exercise and nutrition prehabilitation on key postoperative outcomes, including length of stay, complications, and quality-of-life in various surgical contexts. Overall, this study aims to provide a generalizable understanding of prehabilitation's utility and how to optimize recovery outcomes.

Methods

This systematic review and meta-analysis were conducted and reported in accordance with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-analyses) guidelines.

Search strategy

Two authors (CTC, AEH) independently searched MEDLINE, Cochrane, EMBASE, Web of Science, and Google Scholar from March 2004 to March 2024, utilizing terms such as “surgical”, “prehabilitation”, “exercise”, and “nutrition” (**Supplemental Digital Content 1 and 2**, <http://links.lww.com/JACS/A649>). Abstracts were independently and blindly screened by two authors (CTC, AEH) for randomized controlled trials (RCTs) assessing exercise and nutrition-based interventions in surgical patients. A third reviewer (NHN) resolved any disagreements during the review process.

Selection Criteria

Articles were considered for full-text review if the following inclusion criteria were met: a RCT with a sample size of over 45 adult patients, all at least 18 years old, involving a preoperative exercise or nutrition-based intervention, and measuring relevant postoperative outcomes such as length of stay, complications, pain, mental health, and quality of life. An exercise-based intervention was defined as any program involving supervised or unsupervised physical activity, including high-intensity training, stretching, weightlifting, cardio, and resistance training for at least one week prior to surgery. A nutrition-based intervention was defined as any preoperative nutrition program that aimed at increasing nutritional reserves through oral supplementation and high-protein meals for more than one day before surgery. Studies were excluded if they involved pediatric patients, did not include a preoperative intervention (exercise or nutrition), did not report outcomes of interest, did not have a sample of 45 patients, or were published in a non-English language.

Outcomes

Two primary outcomes were assessed. The first primary outcome was the total length of hospital stay (LOS), defined as the number of days from the day of hospital admission to the day of discharge. The second primary outcome was any postoperative complication within a 12-week postoperative period. Such complications included: urinary tract infection, acute kidney injury, acute urinary retention, delirium, syncope or collapse, cardiac arrest, myocardial infarction, red blood cell transfusion, hematoma, new-onset atrial fibrillation, stroke or transient ischemic attack, surgical site infection or sepsis, anastomotic leak, postoperative hemorrhage, paralytic or mechanical ileus, intra-abdominal abscess, deep vein thrombosis, liver insufficiency, diabetic decompensation, pulmonary embolism, pneumonia, and all-cause mortality. Secondary outcomes included patient-reported outcomes such as QoL, pain, and mental health.

Quality assessment

The Cochrane Risk of Bias tool 2.0 (RoB 2.0) was used by two independent reviewers (CTC and JMC) to assess the quality of the RCTs.²² This tool assesses bias based on five domains: randomization process, deviations from intended intervention, missing outcome, measure of the outcome, and selection of the reported result. Each domain is labeled as “low risk of bias,” “some concerns,” or “high risk of bias” based on responses to signaling questions and a given algorithm. Additionally, publication bias was visually assessed using funnel plots to detect asymmetry, which could indicate selective reporting or an uneven distribution of study results.

Statistical analysis

Statistical analyses were performed using RStudio (Posit, Boston, MA) with “metafor,” “meta,” “netmeta,” and “dmetar” packages.²³ Random effects meta-analysis was performed and statistical heterogeneity was assessed using I^2 statistics. Continuous variables between the prehabilitation and control group were reported as mean difference (MD) when outcome

measurements used the same scale and 95% confidence interval (CI).²⁴ Patient-reported quality-of-life, pain, and mental health outcomes were reported using different scales, and therefore, a standardized mean difference (SMD) was calculated.²⁴ Categorical variables between the prehabilitation group and control group were expressed as odds ratios (OR) and 95% CI. A meta-analysis, meta-proportion, and network meta-analysis were utilized to compare exercise or nutrition, exercise versus nutrition, and exercise and nutrition individually.

Results

Of 298 studies screened, 23 RCTs^{11,13,25–45} comprising 2,182 participants (1,100 males) were included in the meta-analysis (**Supplemental Digital Content 3 and 4**, <http://links.lww.com/JACS/A649>). 1,082 (49.6%) participants received prehabilitation (exercise or nutrition), while 1,100 (50.4%) received standard treatment or placebo. 34.8% of studies were conducted in participants undergoing total knee or hip arthroplasty, 21.7% underwent cardiac surgery, 21.7% underwent abdominal surgery, and 8.7% underwent spine surgery. Across prehabilitation programs, 78.3% (18 studies) involved exercise interventions, and 21.7% (5 studies) utilized nutrition-based approaches (**Table 1**). Exercise-based prehabilitation was assessed across various surgery types, including 44.4% orthopedic, 16.7% abdominal, and 16.7% cardiac. In comparison, nutrition-based prehabilitation was mainly evaluated in participants undergoing gastrointestinal and cardiac surgery.

Among nutrition-based prehabilitation programs (5 studies), 60.0% used commercially available Impact Advanced Recovery solution, 20.0% provided daily meals, and the remaining provided a unique dietary supplement ranging in protein, carbohydrate, arginine, and vitamin concentration. Among exercise-based programs (18 studies), 61.1% involved strength and resistance training, while the remaining studies used high-intensity training (27.8%) or low/moderate intensity training (11.1%).

Bias

The risk of bias assessment showed that most studies included in the meta-analysis were either low risk or had some concerns across the evaluated domains. Areas of concern included missing outcome data and deviations from intended interventions (**Supplemental Digital Content 5**, <http://links.lww.com/JACS/A649>). For example, Liang et al. noted that, due to the project's design, patients, the research team, and prehabilitation staff could not be blinded to the intervention, which could lead to experimenter bias³³. Additionally, a few studies had a high risk of bias because of the selection of the reported results. Rooks et al. reported their primary outcome, pain, at various time points, providing means and standard deviations but not sample sizes²⁶. Similarly, Tew et al. reported SF-36 mental health scores with means and standard deviations at baseline, but subsequent postoperative time points included only the mean score and the difference⁴⁰. Overall, there remains high-quality evidence across most studies.

Effect of Exercise or Nutrition-based Prehabilitation on Surgical Outcomes

We first sought to elucidate the overall impact of nutrition- or exercise-based prehabilitation using a meta-analysis. LOS and complications were assessed due to consistent reporting across studies. Patient-reported outcomes could not be assessed due to insufficient nutrition-based prehabilitation studies reporting such outcomes. Thus, patient-reported outcomes were examined only for the exercise-based prehabilitation studies described below.

For LOS, 17 studies representing 1,546 total participants (n = 776 participants receiving prehabilitation) were analyzed (**Figure 1A; Supplemental Digital Content 5A**, <http://links.lww.com/JACS/A649>). A random effects model found that prehabilitation reduces LOS (MD -0.44 days, 95% CI:-0.78,-0.11, p=0.01, $I^2=44.8\%$) compared to standard treatment. For complications, 14 studies encompassing 1,287 participants (n = 649 participants as part of the prehabilitation intervention group) were included (**Figure 1B, Supplemental Digital**

Content 5B, <http://links.lww.com/JACS/A649>). Prehabilitation was found to reduce complications (OR 0.52, 95% CI:0.35,0.78, $p<0.002$, $I^2=47.1\%$) compared to standard treatment. Taken together, these data suggest that either exercise or nutrition can improve LOS and reduce postoperative complications.

Effect of Exercise Only Prehabilitation on Surgical Outcomes

To better understand the individual role of each intervention type on surgical outcomes, a meta-analysis was conducted for each intervention in isolation. Exercise only trended toward reducing LOS (MD -0.21 days, 95%CI= -0.51,0.09, $p=0.18$, $I^2=59\%$) compared to standard treatment (**Figure 2A; Supplemental Digital Content 6A**, <http://links.lww.com/JACS/A649>). Given the moderate heterogeneity in the meta-analysis, we performed a meta-regression controlling for surgery type, estimated total intervention time (minutes), and exercise type. These moderators were selected due to their observed variability and heterogeneity across prior studies. Moderators were assessed independently and in combination but were not found to predict LOS. Next, total complications were assessed. Unlike LOS, exercise only was found to reduce the odds of any postoperative complication (OR 0.45, 95% CI: 0.28,0.74, $p<0.002$, $I^2=41.2\%$) compared to standard treatment (**Figure 3A; Supplemental Digital Content 6B**, <http://links.lww.com/JACS/A649>).

Effect of Nutrition-Only Prehabilitation on Surgical and Patient-Reported Outcomes

Similarly, we evaluated the effect of nutrition only prehabilitation on LOS and complications. Nutrition-based prehabilitation was associated with a shorter LOS than the standard treatment (MD -1.09 days, 95% CI: -1.72,-0.47, $p<0.001$, $I^2=0\%$) (**Figure 2B; Supplemental Digital Content 7A**, <http://links.lww.com/JACS/A649>). A meta-regression was subsequently deployed to identify predictors for improved LOS. Surgery type and intervention length were not significant predictors of LOS. In contrast, nutrition only was not found to reduce

complications (OR 0.66, 95% CI: 0.34,1.28, $p=0.22$, $I^2=53.9\%$) compared with standard treatment (**Figure 3B; Supplemental Digital Content 7B**, <http://links.lww.com/JACS/A649>).

Exercise versus Nutrition-based Prehabilitation on Surgical Outcomes

While our total cohort meta-analysis demonstrated improvements in both primary outcomes, our subgroup analysis suggested that the two interventions differentially impacted LOS and complications. Two analyses were conducted to compare the effect of each intervention on LOS and complications: a meta-proportion and network meta-analysis. In comparing the interventions, a meta-proportion showed that nutrition induced a greater reduction in LOS compared to exercise (MD -1.09 days, 95% CI: -1.72, -0.47 vs. MD -0.21 days, 95% CI: 0.51, 0.09; $p=0.01$). These results were supported by our network meta-analysis, which showed similar results; nutrition intervention was found to more significantly reduce LOS compared to exercise (MD -0.89 days, 95% CI: -1.62, -0.18, $p=0.01$).

In contrast, no differences were observed in complications between the intervention groups (OR 0.45, 95% CI: 0.28, 0.74 vs. OR 0.66, 95% CI: 0.34, 1.28; $p=0.36$). Similarly, network meta-analysis did not significantly display any differences between exercise and nutrition. Ultimately, prehabilitation modalities may offer unique advantages and affect surgical outcomes to different extents.

Effect of Exercise-based Prehabilitation on Patient-Reported Outcomes

61.1% of exercise-based prehabilitation reported outcomes of QoL, pain and mental health. Sufficient data was provided to conduct a meta-analysis and meta-regression. Overall, QoL increased in those who underwent exercise-based prehabilitation compared to standard treatment.

Quality-of-Life

Of the 18 exercise-based prehabilitation studies, 27.8% reported QoL outcome measures using VAS (Visual Analog Scale) European Quality of Life 5 Dimension Health Questionnaire (EQ5D), and MacNew QoL. Most studies (80.0%) utilized VAS EQ5D. Meta-analysis employing a random effects model found that exercise-based prehabilitation improved QoL (SMD 0.94, 95% CI: 0.24,1.64, $p=0.01$, $I^2=93.6\%$) (**Figure 4A; Supplemental Digital Content 8A**, <http://links.lww.com/JACS/A649>). No significant predictors that impacted QoL were identified.

Pain

27.8% of exercise-based prehabilitation studies reported longitudinal pain scores using VAS or the Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC). A random effects model assessed the SMD between intervention and control groups. Exercise-based prehabilitation was not found to reduce reported pain significantly (SMD -2.49, 95% CI: -7.58,2.60, $p=0.34$, $I^2=97.6\%$) (**Figure 4B; Supplemental Digital Content 8B**, <http://links.lww.com/JACS/A649>). Further investigation of the survey type, total intervention length, or surgery type yielded no significant moderators.

Mental Health

Lastly, five studies reported on mental health using a variety of instruments, including the Beck Depression Index (BDI), Hospital Anxiety and Depression Scale (HADS), and Short Form-36 (SF36) Mental Component Summary (MCS) or Mental Health (MH) Score. 60.0% of studies utilized the SF36 to assess mental health, while the remaining studies each utilized HADS or BDI. To assess the impact of exercise-based prehabilitation, we evaluated the change in postoperative compared to preoperative scores between intervention and control groups using standardized mean differences. A meta-analysis found that the intervention group trended

towards improved mental health scores (SMD 2.44, 95% CI: -1.45,6.32, $p=0.22$, $I^2=98.2\%$) (Figure 4C; Supplemental Digital Content 8C, <http://links.lww.com/JACS/A649>). A meta-regression was performed to address the heterogeneity using potential moderators; however, neither surgery category nor outcome measure aggregation impacted the model.

Discussion

Exercise and nutrition-based prehabilitation programs have shown promise to optimize surgical patients and improve postoperative outcomes and recovery. However, despite growing interest, prior studies of prehabilitation have been narrow in scope either by surgery or intervention type and report mixed findings, limiting the extent to which conclusions can be drawn regarding their impact on overall surgical outcomes. In the current work, we found that either exercise or nutrition in prehabilitation protocols improved length of stay and complications after surgery. However, when subgroup analysis for each intervention was performed, exercise-based prehabilitation reduced complications and not LOS. Conversely, nutrition-based prehabilitation reduced LOS but not complications. When we compared the interventions against each other, nutrition-based prehabilitation reduced LOS more than exercise-based prehabilitation. While we did not see a difference between the interventions on complications in our meta-proportion analysis, it is possible that the substantial difference between surgery type may have influenced these findings as the majority of exercise-based prehabilitation focused on orthopaedic surgeries whereas nutrition-based studies were predominantly gastrointestinal and cardiac surgeries. Our work demonstrates that exercise or nutritional prehabilitation improve surgical outcomes, albeit the two different modalities contribute to the improvement of different outcomes.

A key consideration for prehabilitation is how to integrate it into current practice. Practically, it is clear that a certain amount of investment up front will be necessary to design the appropriate protocol for the index surgery as different patient populations for different indications (e.g. cancer vs. orthopaedic) will likely have different prehabilitation needs. However, an emerging body of literature, including our own work on a multimodal prehabilitation protocol prior to facial reconstruction, has demonstrated that the incorporation of technology for virtual and/or artificial intelligence-assisted delivery is both feasible and effective.⁴⁶⁻⁴⁸ Additionally, a secondary immediate benefit that we have noticed from our experience is that the implementation of the virtual prehabilitation protocol correlated with a reduction in patient questions through the patient portal to our clinic staff in both the preoperative and postoperative period. While the exact reason for the decreased patient messages could not be elucidated, it is possible that the increased and readily accessible information improved patient readiness for surgery and recovery.

The observation of clear benefits for both interventions may be practically interpreted to imply the value of multimodal prehabilitation approaches. Prior studies of prehabilitation have traditionally focused on programs designed to deliver one or the other interventions (i.e. exercise or nutrition), curated in a manner influenced by the specific surgical setting under study. For instance, exercise-based prehabilitation has been primarily explored in the context of orthopedic or cardiothoracic surgery^{11,26,30,36,37,43,49}, while nutrition-based prehabilitation has been predominantly applied to gastrointestinal surgery^{38,39,50}. Yet, a growing body of evidence suggests that multimodal prehabilitation programs may improve outcomes to a greater magnitude than any single intervention.⁵¹ Our current work illustrated that distinct benefits are associated with different prehabilitation modalities, suggesting varied mechanisms underlying their effects and how they manifest as differential outcomes. There may be several explanations for how each

prehabilitation modality can influence outcomes. Surgery has been shown to promote inflammation and increase oxygen demand, potentially leading to muscle catabolism^{52,53}. Additionally, the combined effects of anesthesia, pain, and postoperative immobility may compromise inspiratory muscle strength⁵⁴. The effects of exercise can address these postoperative physiologic issues, including reducing oxidative stress and inflammatory markers^{55,56}. Studies have shown how malnutrition can lead to weakened immune systems, impaired wound healing, or compromised digestive function⁵⁷⁻⁵⁹. Prehabilitation focused on nutrition can optimize aspects of postoperative recovery and potentially reduce length of stay. However, it is still unclear which component of a multimodal intervention approach contributes the most to improved surgical outcomes. Additional research in this area would help clarify their specific impacts.

Taken together, our findings support the value of prehabilitation programs and suggest their potential for broader integration into routine perioperative care. However, prehabilitation studies may benefit from greater standardization and the development of evidence-based guidelines for intervention characteristics. For example, future efforts may be beneficially informed by data clarifying the impact of duration, intensity, and type of exercise and by more clearly defined nutritional supplementation goals. In our review, most exercise-based studies utilized resistance training or high-intensity intervals, while others used posture training and stretching. Furthermore, the duration of the intervention varied from 2 weeks to 24 weeks. These differences may further be exacerbated by varying compliance levels stemming from a lack of supervision, implying the importance of personalized prehabilitation, wherein interventions are tailored according to patient-specific and procedure-specific risk factors such as frailty and surgical complexity. For instance, risk-profile-informed prehabilitation may entail prioritization of nutrition-focused interventions for malnourished patients, while structured exercise regimens

may be implemented specifically in frail or vascularly-compromised patients at highest risk for complications. Such personalized approaches may also enable programmed adaptability, allowing for individualized adjustment according to patient progress. Further clarification of cost effects and long-term outcomes may help expedite integration into existing healthcare infrastructures.

Beyond surgical outcomes, prehabilitation may impact patient-reported psychosocial functioning. Our study found a significant increase in QoL in patients who underwent exercise-based prehabilitation, suggesting improved overall well-being and function. However, the absence of impacts on pain or mental health outcomes suggests alternative interventions—such as psychosocial interventions employed elsewhere in prehabilitation programs—may be more suitable for addressing psychological or pain-related aspects of surgical recovery.⁶⁰ Psychology has been shown in the literature to influence surgical outcomes across most surgical subspecialties.^{6,61} Sobol-Swapinska et al. found in a systematic review and meta-analysis that surgical patients with positive expectations and positive outlook before surgery report lower levels of acute postsurgical pain.⁶² Poorly controlled postoperative pain has been previously associated with impaired QoL and prolonged recovery.⁶³ As such, prehabilitation programs may benefit from not only exercise and nutrition but also psychological interventions like cognitive behavioral therapy which is known to change destructive or disturbing thought patterns and improve the ability to cope with stress.⁶⁴ Ultimately, integrating various prehabilitation strategies into a unified multimodal intervention may help optimize different outcomes, as some interventions may have a greater impact on specific outcomes than others.

This review has limitations worth considering. Our search criteria focused on prehabilitation, likely excluding more nutrition-based preoperative interventions that weren't labeled as such, like ERAS nutrition studies. This could explain the three-fold fewer nutrition

studies compared to exercise approaches. Our strict criteria further limited eligible studies from the past 20 years. Additionally, variability in nutrition and exercise protocols may affect outcome measures, as moderate heterogeneity was observed in nutritional interventions and surgical outcome reporting ($I^2=38.7-53.9\%$). For instance, nutrition protocols varied from regimented supplementation to precooked meals without supplements, influencing observed effects. Standardized intervention characteristics and outcome measures in future research would help strengthen generalizability and facilitate systematic reviews and meta-analyses across surgical subspecialties. As with other reviews, significant heterogeneity is a limitation and can be addressed by investigating moderators like body mass index, frailty score, and comorbidities that may affect LOS and complications.

Conclusion

In this systematic review and meta-analysis, pooled results indicate several positive associations between nutrition and exercise-based prehabilitation interventions and postsurgical outcomes compared to standard treatment. These benefits lend support to integrating structured prehabilitation into standard perioperative care. Exercise-based programs tend to be more effective at reducing complications, while nutrition-centered approaches contribute more significantly to reduced LOS. These findings suggest that preoperative exercise programs and nutritional supplementation differentially improve patient outcomes across diverse surgical populations and that prehabilitation programs may broadly serve increasingly significant roles as the landscape of surgical care evolves.

Appendix

Members of the Craniofacial Outcomes Research Team include Youngnam Kang, PhD, Wei Chen, PhD, Archi K Patel, and Xiaoyan Ren, MD, PhD.

Division of Plastic and Reconstructive Surgery, Department of Surgery, University of California Los Angeles, David Geffen School of Medicine and the Greater Los Angeles VA Healthcare System, Los Angeles, CA

ACCEPTED

References

1. Etzioni DA, Liu JH, Maggard MA, Ko CY. The Aging Population and Its Impact on the Surgery Workforce. *Ann Surg*. 2003;238(2):170. doi:10.1097/01.SLA.0000081085.98792.3d
2. Meara JG, Leather AJM, Hagander L, et al. Global Surgery 2030: evidence and solutions for achieving health, welfare, and economic development. *Int J Obstet Anesth*. 2016;25:75-78. doi:10.1016/j.ijoa.2015.09.006
3. Healey MA, Shackford SR, Osler TM, Rogers FB, Burns E. Complications in Surgical Patients. *Arch Surg*. 2002;137(5):611-618. doi:10.1001/archsurg.137.5.611
4. Looijaard SMLM, Slee-Valentijn MS, Otten RHJ, Maier AB. Physical and Nutritional Prehabilitation in Older Patients With Colorectal Carcinoma: A Systematic Review. *J Geriatr Phys Ther*. 2018;41(4):236. doi:10.1519/JPT.0000000000000125
5. Clark MA, Plank LD, Hill GL. Wound healing associated with severe surgical illness. *World J Surg*. 2000;24(6):648-654. doi:10.1007/s002689910106
6. Paredes AZ, Hyer JM, Diaz A, Tsilimigras DI, Pawlik TM. The Impact of Mental Illness on Postoperative Outcomes... : *Annals of Surgery*. 2020;272(3):419-425. doi:10.1097/SLA.00000000000004118
7. Steffens D, Beckenkamp PR, Hancock M, Solomon M, Young J. Preoperative exercise halves the postoperative complication rate in patients with lung cancer: a systematic review of the effect of exercise on complications, length of stay and quality of life in patients with cancer. *Br J Sports Med*. 2018;52(5):344. doi:10.1136/bjsports-2017-098032
8. Steffens D, Beckenkamp PR, Young J, Solomon M, da Silva TM, Hancock MJ. Is preoperative physical activity level of patients undergoing cancer surgery associated with postoperative outcomes? A systematic review and meta-analysis. *Eur J Surg Oncol*. 2019;45(4):510-518. doi:10.1016/j.ejso.2018.10.063

9. Evans DC, Martindale RG, Kiraly LN, Jones CM. Nutrition Optimization Prior to Surgery. *Nutr Clin Pract*. 2014;29(1):10-21. doi:10.1177/0884533613517006
10. Lobo DN, Pavel Skořepa, Gomez D, Greenhaff PL. Prehabilitation: high-quality evidence is still required. *Br J Anaesth*. 2023;130(1):9-14. doi:10.1016/j.bja.2022.09.016
11. Beaupre LA, Lier D, Davies DM, Johnston DBC. The effect of a preoperative exercise and education program on functional recovery, health related quality of life, and health service utilization following primary total knee arthroplasty. *J Rheumatol*. 2004;31(6):1166-1173.
12. Whitney JAD, Parkman S. Preoperative physical activity, anesthesia, and analgesia: Effects on early postoperative walking after total hip replacement. *Appl Nurs Res*. 2002;15(1):19-27. doi:10.1053/apnr.2002.29519
13. Nielsen PR, Jørgensen LD, Dahl B, Pedersen T, Tønnesen H. Prehabilitation and early rehabilitation after spinal surgery: randomized clinical trial. *Clin Rehabil*. 2010;24(2):137-148. doi:10.1177/0269215509347432
14. Bozzetti F, Gianotti L, Braga M, Di Carlo V, Mariani L. Postoperative complications in gastrointestinal cancer patients: the joint role of the nutritional status and the nutritional support. *Clin Nutr Edinb Scotl*. 2007;26(6):698-709. doi:10.1016/j.clnu.2007.06.009
15. Gustafsson UO, Scott MJ, Hubner M, et al. Guidelines for Perioperative Care in Elective Colorectal Surgery: Enhanced Recovery After Surgery (ERAS®) Society Recommendations: 2018. *World J Surg*. 2019;43(3):659-695. doi:10.1007/s00268-018-4844-y
16. Braga M, Ljungqvist O, Soeters P, Fearon K, Weimann A, Bozzetti F. ESPEN Guidelines on Parenteral Nutrition: Surgery. *Clin Nutr*. 2009;28(4):378-386. doi:10.1016/j.clnu.2009.04.002
17. McIsaac DI, Gill M, Boland L, et al. Prehabilitation in adult patients undergoing surgery: an umbrella review of systematic reviews. *Br J Anaesth*. 2022;128(2):244-257. doi:10.1016/j.bja.2021.11.014

18. Amirkhosravi F, Allenson KC, Moore LW, et al. Multimodal prehabilitation and postoperative outcomes in upper abdominal surgery: systematic review and meta-analysis. *Sci Rep*. 2024;14(1):16012. doi:10.1038/s41598-024-66633-6
19. Zhou L, Li H, Zhang Z, Wang L. Effects of multimodal prehabilitation and exercise prehabilitation on patients undergoing colorectal surgery: A systematic review and meta-analysis of randomised controlled trials. *J Glob Health*. 2024;14:04239. doi:10.7189/jogh.14.04239
20. Sowerbutts AM, Burden S, Sremanakova J, French C, Knight SR, Harrison EM. Preoperative nutrition therapy in people undergoing gastrointestinal surgery - Sowerbutts, AM - 2024 | Cochrane Library. Accessed December 16, 2024. <https://www.cochranelibrary.com/cdsr/doi/10.1002/14651858.CD008879.pub3/full>
21. Burden S, Todd C, Hill J, Lal S. Pre-operative Nutrition Support in Patients Undergoing Gastrointestinal Surgery. - Burden, S - 2012 | Cochrane Library. Accessed December 16, 2024. <https://www.cochranelibrary.com/cdsr/doi/10.1002/14651858.CD008879.pub2/abstract>
22. Sterne JAC, Savović J, Page MJ, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ*. 2019;366:l4898. doi:10.1136/bmj.l4898
23. Citing RStudio. Posit Support. April 26, 2024. Accessed November 5, 2024. <https://support.posit.co/hc/en-us/articles/206212048-Citing-RStudio>
24. Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). Cochrane Handbook for Systematic Reviews of Interventions version 6.5 (updated August 2024). Cochrane, 2024. Available from www.training.cochrane.org/handbook.
25. Berkel AEM, Bongers BC, Kotte H, et al. Effects of Community-based Exercise Prehabilitation for Patients Scheduled for Colorectal Surgery With High Risk for Postoperative Complications: Results of a Randomized Clinical Trial. *Ann Surg*. 2022;275(2):e299-e306. doi:10.1097/SLA.0000000000004702

26. Rooks DS, Huang J, Bierbaum BE, et al. Effect of preoperative exercise on measures of functional status in men and women undergoing total hip and knee arthroplasty. *Arthritis Rheum.* 2006;55(5):700-708. doi:10.1002/art.22223
27. Marchand AA, Houle M, O'Shaughnessy J, Châtillon CÉ, Cantin V, Descarreaux M. Effectiveness of an exercise-based prehabilitation program for patients awaiting surgery for lumbar spinal stenosis: a randomized clinical trial. *Sci Rep.* 2021;11(1):11080. doi:10.1038/s41598-021-90537-4
28. Norouzi M, Nadjarzadeh A, Maleki M, et al. Evaluation of the recovery after heart surgery following preoperative supplementation with a combination of beta-hydroxy-beta-methylbutyrate, L-arginine, and L-glutamine: a double-blind randomized placebo-controlled clinical trial. *Trials.* 2022;23(1):649. doi:10.1186/s13063-022-06621-1
29. Rosenfeldt F, Braun L, Spitzer O, et al. Physical conditioning and mental stress reduction--a randomised trial in patients undergoing cardiac surgery. *BMC Complement Altern Med.* 2011;11:20. doi:10.1186/1472-6882-11-20
30. Akowuah EF, Wagnild JM, Bardgett M, et al. A randomised controlled trial of prehabilitation in patients undergoing elective cardiac surgery. *Anaesthesia.* 2023;78(9):1120-1128. doi:10.1111/anae.16072
31. IJmker-Hemink VE, Wanten GJA, de Nes LCF, van den Berg MGA. Effect of a Preoperative Home-Delivered, Protein-Rich Meal Service to Improve Protein Intake in Surgical Patients: A Randomized Controlled Trial. *JPEN J Parenter Enteral Nutr.* 2021;45(3):479-489. doi:10.1002/jpen.2015
32. Tepaske R, te Velthuis H, Oudemans-van Straaten HM, et al. Glycine does not add to the beneficial effects of perioperative oral immune-enhancing nutrition supplements in high-risk

cardiac surgery patients. *JPEN J Parenter Enteral Nutr.* 2007;31(3):173-180.

doi:10.1177/0148607107031003173

33. Liang MK, Bernardi K, Holihan JL, et al. Modifying Risks in Ventral Hernia Patients With Prehabilitation: A Randomized Controlled Trial. *Ann Surg.* 2018;268(4):674-680.

doi:10.1097/SLA.0000000000002961

34. Barberan-Garcia A, Ubré M, Roca J, et al. Personalised Prehabilitation in High-risk Patients Undergoing Elective Major Abdominal Surgery: A Randomized Blinded Controlled Trial. *Ann Surg.* 2018;267(1):50-56. doi:10.1097/SLA.0000000000002293

35. Villadsen A, Overgaard S, Holsgaard-Larsen A, Christensen R, Roos EM. Postoperative effects of neuromuscular exercise prior to hip or knee arthroplasty: a randomised controlled trial. *Ann Rheum Dis.* 2014;73(6):1130-1137. doi:10.1136/annrheumdis-2012-203135

36. Bitterli R, Sieben JM, Hartmann M, de Bruin ED. Pre-surgical sensorimotor training for patients undergoing total hip replacement: a randomised controlled trial. *Int J Sports Med.* 2011;32(9):725-732. doi:10.1055/s-0031-1271696

37. Steinmetz C, Bjarnason-Wehrens B, Baumgarten H, Walther T, Mengden T, Walther C. Prehabilitation in patients awaiting elective coronary artery bypass graft surgery - effects on functional capacity and quality of life: a randomized controlled trial. *Clin Rehabil.*

2020;34(10):1256-1267. doi:10.1177/0269215520933950

38. Barker LA, Gray C, Wilson L, Thomson BNJ, Shedda S, Crowe TC. Preoperative immunonutrition and its effect on postoperative outcomes in well-nourished and malnourished gastrointestinal surgery patients: a randomised controlled trial. *Eur J Clin Nutr.* 2013;67(8):802-807. doi:10.1038/ejcn.2013.117

39. Hübner M, Cerantola Y, Grass F, Bertrand PC, Schäfer M, Demartines N. Preoperative immunonutrition in patients at nutritional risk: results of a double-blinded randomized clinical trial. *Eur J Clin Nutr*. 2012;66(7):850-855. doi:10.1038/ejcn.2012.53
40. Tew GA, Batterham AM, Colling K, et al. Randomized feasibility trial of high-intensity interval training before elective abdominal aortic aneurysm repair. *Br J Surg*. 2017;104(13):1791-1801. doi:10.1002/bjs.10669
41. Matassi F, Duerinckx J, Vandenneucker H, Bellemans J. Range of motion after total knee arthroplasty: the effect of a preoperative home exercise program. *Knee Surg Sports Traumatol Arthrosc Off J ESSKA*. 2014;22(3):703-709. doi:10.1007/s00167-012-2349-z
42. Woodfield JC, Clifford K, Wilson GA, Munro F, Baldi JC. Short-term high-intensity interval training improves fitness before surgery: A randomized clinical trial. *Scand J Med Sci Sports*. 2022;32(5):856-865. doi:10.1111/sms.14130
43. Topp R, Swank AM, Quesada PM, Nyland J, Malkani A. The effect of prehabilitation exercise on strength and functioning after total knee arthroplasty. *PM R*. 2009;1(8):729-735. doi:10.1016/j.pmrj.2009.06.003
44. Tungtrongjit Y, Weingkum P, Saunkool P. The effect of preoperative quadriceps exercise on functional outcome after total knee arthroplasty. *J Med Assoc Thail Chotmai het Thangphaet*. 2012;95 Suppl 10:S58-66.
45. Blasco JM, Acosta-Ballester Y, Martínez-Garrido I, García-Molina P, Igual-Camacho C, Roig-Casasús S. The effects of preoperative balance training on balance and functional outcome after total knee replacement: a randomized controlled trial. *Clin Rehabil*. 2020;34(2):182-193. doi:10.1177/0269215519880936
46. Hall AE, Nguyen NH, Patel AK, et al. Development, Implementation, and Evaluation of The Prehabilitation for Facial Feminization Surgery (PREFACE) Program. *Submiss*.

47. Moya A, Keane C, Chen Y, et al. Feasibility of a virtual multimodal prehabilitation intervention for patients with cancer undergoing surgery. *Clin Nutr ESPEN*. 2025;66:121-134. doi:10.1016/j.clnesp.2025.01.024
48. Talen AD, Leenen JPL, van der Sluis G, Oldenhuis HKE, Klaase JM, Patijn GA. Feasibility of a Comprehensive eCoach to Support Patients Undergoing Colorectal Surgery: Longitudinal Observational Study. *JMIR Perioper Med*. 2025;8:e67425. doi:10.2196/67425
49. McIsaac DI, Hladkiewicz E, Bryson GL, et al. Home-based prehabilitation with exercise to improve postoperative recovery for older adults with frailty having cancer surgery: the PREHAB randomised clinical trial. *Br J Anaesth*. 2022;129(1):41-48. doi:10.1016/j.bja.2022.04.006
50. Giger-Pabst U, Lange J, Maurer C, et al. Short-term preoperative supplementation of an immunoenriched diet does not improve clinical outcome in well-nourished patients undergoing abdominal cancer surgery. *Nutr Burbank Los Angel Cty Calif*. 2013;29(5):724-729. doi:10.1016/j.nut.2012.10.007
51. Wijma AG, Hoogwater FJH, Nijkamp MW, Klaase JM. Personalized multimodal prehabilitation reduces cardiopulmonary complications after pancreatoduodenectomy: results of a propensity score matching analysis. *HPB*. 2023;25(11):1429-1437. doi:10.1016/j.hpb.2023.07.899
52. Straaten HMO van, Jansen PGM, te velhuis H, et al. Increased oxygen consumption after cardiac surgery is associated with the inflammatory response to endotoxemia. *Intensive Care Med*. 1996;22(4):294-300. doi:10.1007/BF01700449
53. Schricker T, Gougeon R, Eberhart L, et al. Type 2 diabetes mellitus and the catabolic response to surgery. *Anesthesiology*. 2005;102(2):320-326. doi:10.1097/00000542-200502000-00013

54. Rodrigues AJ, Mendes V, Ferreira PEG, et al. Preoperative respiratory muscle dysfunction is a predictor of prolonged invasive mechanical ventilation in cardiorespiratory complications after heart valve surgery. *Eur J Cardiothorac Surg*. 2011;39(5):662-666. doi:10.1016/j.ejcts.2010.08.021
55. Pierce GL, Donato AJ, LaRocca TJ, Eskurza I, Silver AE, Seals DR. Habitually exercising older men do not demonstrate age-associated vascular endothelial oxidative stress. *Aging Cell*. 2011;10(6):1032-1037. doi:10.1111/j.1474-9726.2011.00748.x
56. Gano LB, Donato AJ, Pierce GL, et al. Increased proinflammatory and oxidant gene expression in circulating mononuclear cells in older adults: amelioration by habitual exercise. *Physiol Genomics*. 2011;43(14):895-902. doi:10.1152/physiolgenomics.00204.2010
57. Bourke CD, Berkley JA, Prendergast AJ. Immune Dysfunction as a Cause and Consequence of Malnutrition. *Trends Immunol*. 2016;37(6):386-398. doi:10.1016/j.it.2016.04.003
58. Stechmiller JK. Understanding the role of nutrition and wound healing. *Nutr Clin Pract Off Publ Am Soc Parenter Enter Nutr*. 2010;25(1):61-68. doi:10.1177/0884533609358997
59. Ward N. Nutrition support to patients undergoing gastrointestinal surgery. *Nutr J*. 2003;2:18. doi:10.1186/1475-2891-2-18
60. Hanley AW, Gililand J, Garland EL. To be mindful of the breath or pain: Comparing two brief preoperative mindfulness techniques for total joint arthroplasty patients. *J Consult Clin Psychol*. 2021;89(7):590-600. doi:10.1037/ccp0000657
61. Josephs CA, Shaffer VO, Kucera WB. Impact of Mental Health on General Surgery Patients and Strategies to Improve Outcomes. *Am Surg*. 2023;89(6):2636-2643. doi:10.1177/00031348221109469

62. Sobol-Kwapinska M, Babel P, Plotek W, Stelcer B. Psychological correlates of acute postsurgical pain: A systematic review and meta-analysis. *Eur J Pain*. 2016;20(10):1573-1586. doi:10.1002/ejp.886
63. Gan TJ. Poorly controlled postoperative pain: prevalence, consequences, and prevention. *J Pain Res*. 2017;10:2287-2298. doi:10.2147/JPR.S144066
64. Nakao M, Shiotsuki K, Sugaya N. Cognitive-behavioral therapy for management of mental health and stress-related disorders: Recent advances in techniques and technologies. *Biopsychosoc Med*. 2021;15(1):16. doi:10.1186/s13030-021-00219-w

Figure Legend

Figure 1. Forest plots of exercise-based or nutrition-based prehabilitation assessing (A) length of stay and (B) complications. MD, mean difference; OR, odds ratio.

Figure 2. Forest plots of prehabilitation compared with standard treatment on length of stay, (A) exercise-based only and (B) nutrition-based only. MD, mean difference.

Figure 3. Forest plots of prehabilitation compared with standard treatment on complications, (A) exercise-based only and (B) nutrition-based only. OR, odds ratio.

Figure 4. Forest plots of the outcomes of exercise compared with standard care on (A) quality-of-life, (B) pain, and (C) mental health. SMD, standard mean difference.

Precis

This systematic review and meta-analysis found that exercise or nutrition-based prehabilitation reduces length of stay and complications across various surgical contexts. However, subgroup analysis revealed that each intervention differentially impacted the primary outcomes.

Table 1: General Characteristics of Included Studies

Study	Country	Protocol type	Operation	Intervention specifics	Protocol description	Intervention Length	Reported outcomes
Akowuah et al, 2023	UK	Exercise	Cardiac surgery	High intensity	2 supervised 60-min sessions/wk	4 wk	LOS, complication, VAS EQ5D, HADS-anxiety
Barberan-Garcia et al, 2018	Spain	Exercise	Abdominal	Aerobic, high intensity, intervals	1 to 3 40-minute sessions/wk	4 wk	LOS, complication
Beaupre et al, 2004	Canada	Exercise	TKA	Resistance training	3 sessions/week per wk	4 wk	LOS, WOMAC, complication, SF-36 Mental Health
Berkel et al, 2022	Netherlands	Exercise	Colorectal	Aerobic, moderate-to-high intensity, intervals, resistance training	3 supervised 60-min sessions/wk	3 wk	LOS, complication
Bitterli et al, 2011	Netherlands	Exercise	THA	Bodyweight movements	6 exercises performed twice daily with repetition scheme	2-6 wk	LOS
Blasco et al, 2020	Spain	Exercise	TKA	Strength training, balance exercises	3 sessions/wk	4 wk	VAS EQ5D
Liang et al, 2018	US	Exercise	Ventral hernia	Aerobic, stretching, resistance training	Weekly physical therapy session	24 wk	LOS, complication

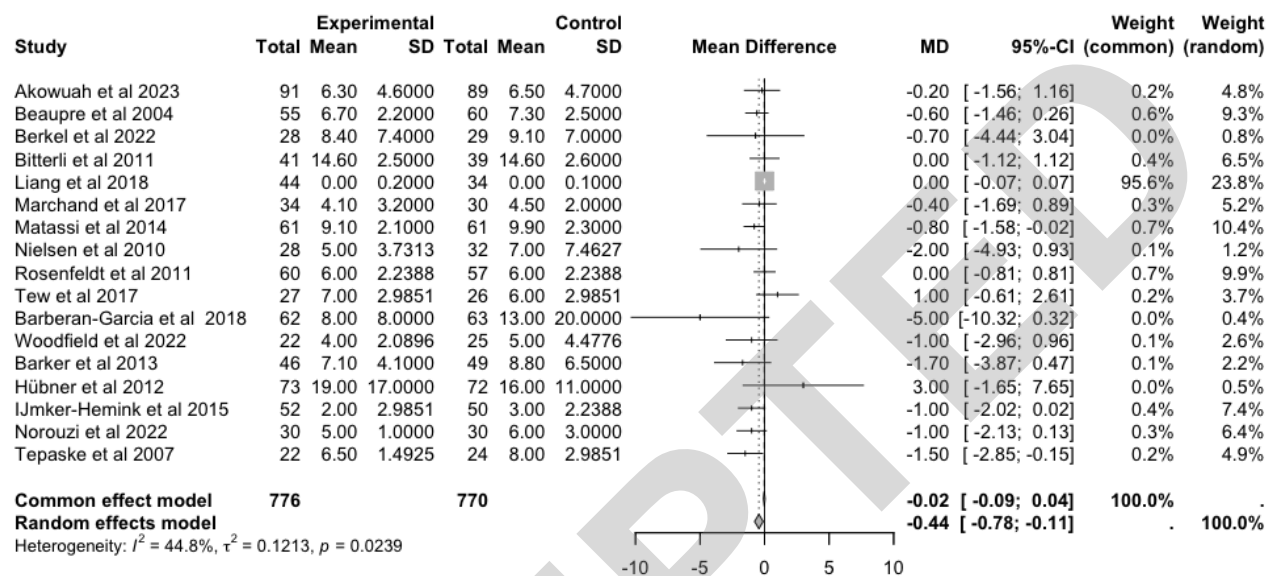
Marchand et al, 2021	Canada	Exercise	Spine	Resistance training	3 d/wk of supervised 25-min sessions	6 wk	LOS, complication
Matassi et al, 2014	Italy	Exercise	TKA	Resistance training, stretching	5 home-based sessions/wk	6 wk	LOS
Nielsen et al, 2010	Denmark	Exercise	Spine	Aerobic, resistance training	30 min daily	6-8 wk	LOS, complication, VAS Pain
Rooks et al, 2006	US	Exercise	TKA	Aerobic, moderate intensity, resistance training, stretching	3 30–60-min sessions/wk	6 wk	WOMAC, complication
Rosenfeldt et al, 2011	Australia	Exercise	CABG	Aerobic, low intensity, continuous	2 60-min sessions/week unsupervised 30-to-60-min aerobic sessions/wk	2 wk	LOS, SF-36 MCS
Steinmetz et al, 2022	Germany	Exercise	CABG	Aerobic, moderate intensity gymnastics	3 d/wk of supervised exercise	2 wk	MacNew QoL
Tew et al, 2017	UK	Exercise	Vascular (AAA)	Aerobic, high intensity, intervals	3 supervised hospital-based sessions/wk	4 wk	LOS, VAS EQ5D, SF36 Mental Health
Topp et al, 2009	US	Exercise	TKA	Resistance training	3 sessions/wk	6-8 wk	VAS Pain
Tungtrongjit et al, 2021	Thailand	Exercise	TKA	Resistance training	3 sessions/d	3 wk	VAS Pain
Villadsen et al, 2014	Denmark	Exercise	THA or TKA	Aerobic, core/posture training resistance training	2 d/wk of 1 h session	8 wk	VAS EQ5D

Woodfield et al, 2022	New Zealand	Exercise	Abdominal	Aerobic, high intensity, intervals	14 supervised 30-min sessions	4 wk	LOS, complication, SF-36 MCS
Barker et al, 2013	Australia	Nutrition	Gastrointestinal	Protein: 18.1 g (arginine: 4.2 g), carbohydrates: 44.7 g, fats: 9.2 g (ALA: 1.1 g, EPA: 672 mg, DHA: 432 mg)	Impact advanced recovery orally 3 times/d	5 d	LOS, complication
Hübner et al, 2012	Switzerland	Nutrition	Gastrointestinal	Protein 16.8 g (arginine: 3.8 g), carbohydrates: 40.2 g, fats: 8.3, (omega-3 FA: 1 g), RNA: 0.45 g	Immune-enhanced nutrition (Impact) orally 3 times/d	5 d	LOS, complication
IJmker-Hemink et al, 2012	Netherlands	Nutrition	Elective*	—	6 protein rich dishes/d to meet 1.2g/kg body weight protein requirement (average 60.8g/d)	3 wk	LOS, complication
Norouzi et al, 2022	Iran	Nutrition	Cardiac surgery	Arginine: 7 g, glutamine: 7 g, β -hydroxy- β -methylbutyrate (HMB): 1.5 g	2 sachets of supplement combination/day	4 wk	LOS, complication
Tepaske et al, 2007	Netherlands	Nutrition	Cardiac surgery	Protein: 8.8 g (whey: 4.35 g, arginine: 1.25 g, glycine: 3.2 g), carbohydrates: 10.2 g, fats: 2.8 g (omega-3 FA: 0.33 g), RNA: 0.13 g	5-10 L of oral Immune-enhanced nutrition (impact); each serving 100 mL	5-10 d	LOS, complication

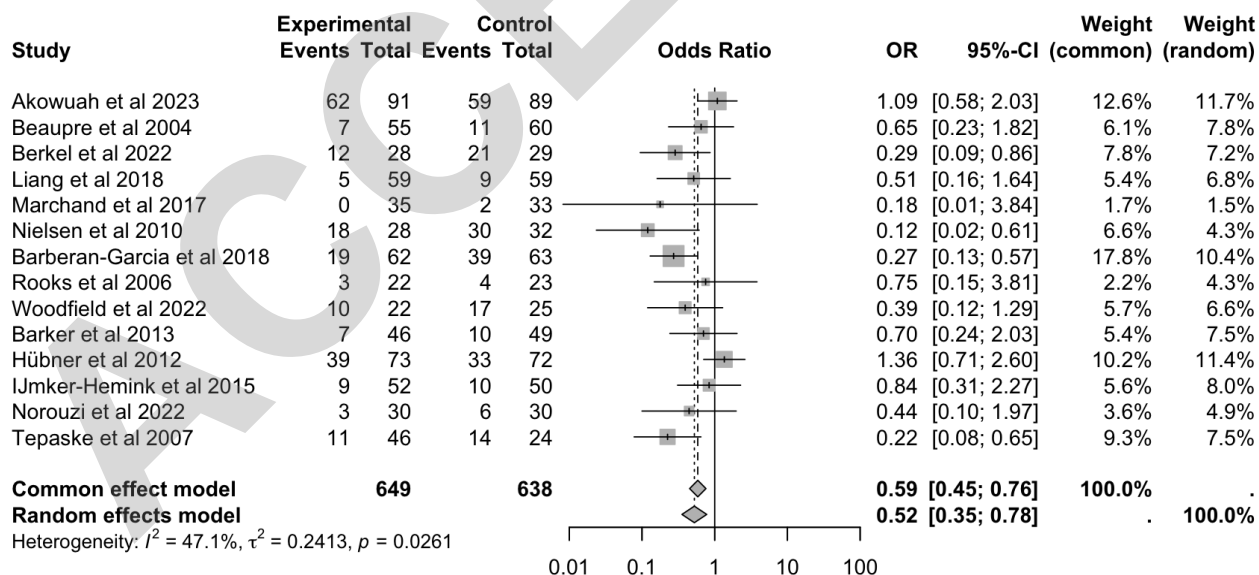
*General surgery, urology, orthopaedic, and gynecology; Intervals: sprint for certain duration at target heart rate, followed by duration of rest or lower intensity; Intensity: determined by percentage of maximum heart rate

AAA, abdominal aortic aneurysm; ALA, alpha-linolenic acid; CABG, coronary artery bypass graft; DHA, docosahexaenoic acid; EPA, eicosapentaenoic acid; EQ5D, European Quality of Life 5 Dimension Health Questionnaire; FA, fatty acid; HADS, Hospital Anxiety and Depression Scale; LOS, length of stay; MCS, mental component summary; QoL, quality of life; sf-63, Short Form-36; TKA, total knee arthroplasty; THA, total hip arthroplasty; VAS, visual analog scale; WOMAC, Western Ontario and McMaster Universities Osteoarthritis Index.

Figure 1

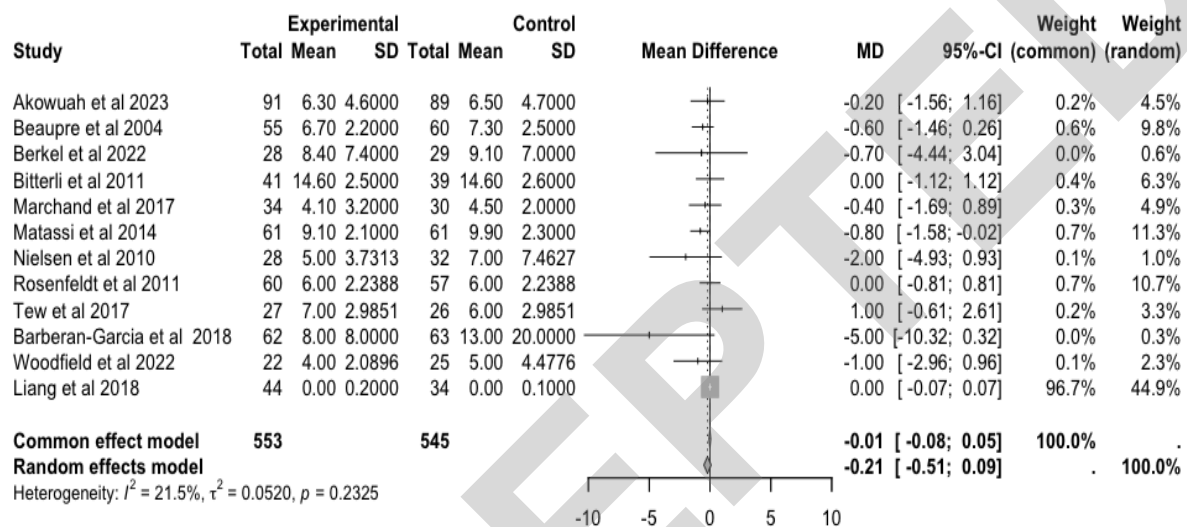


A Length of Stay

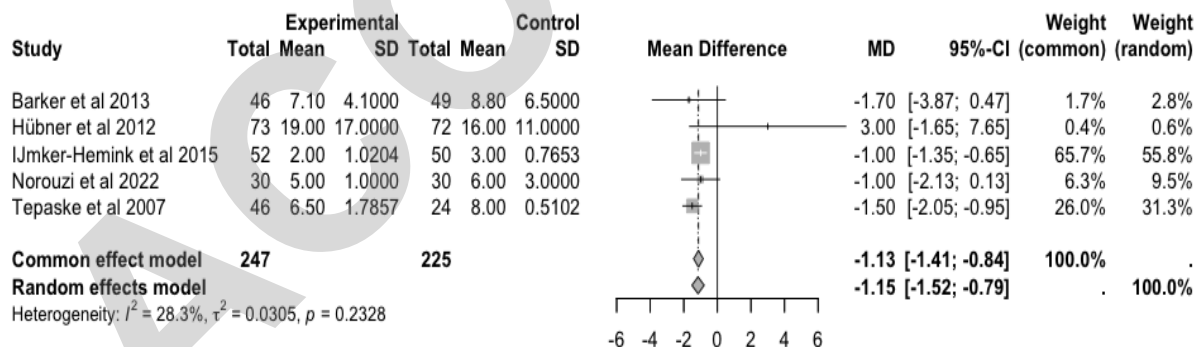


B Complications

Figure 2

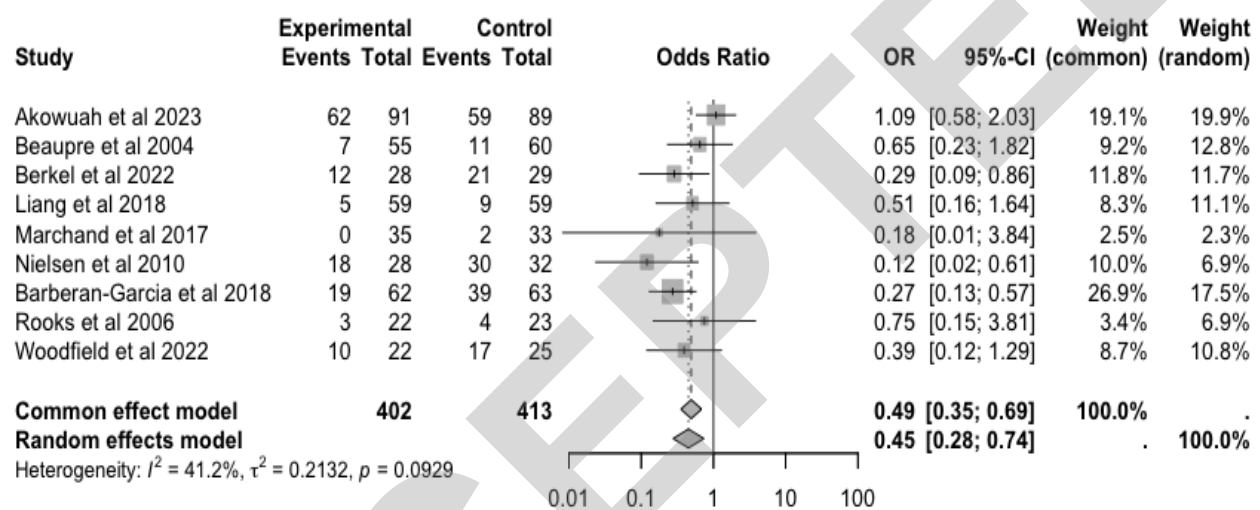


A Exercise-Only Length of Stay

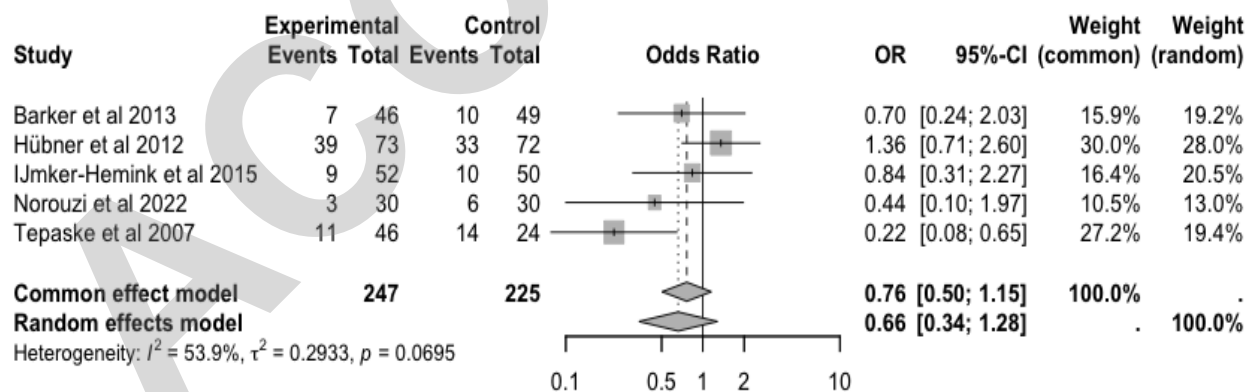


B Nutrition-Only Length of Stay

Figure 3

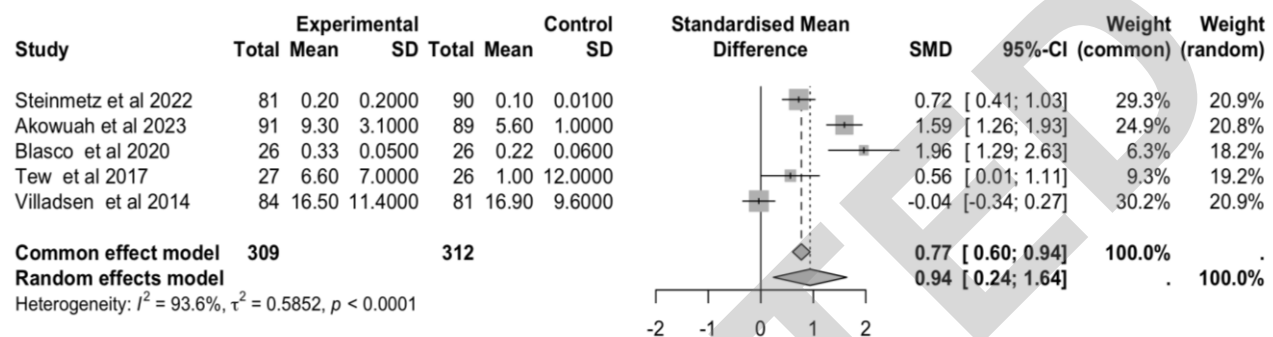


A Exercise-Only Complication

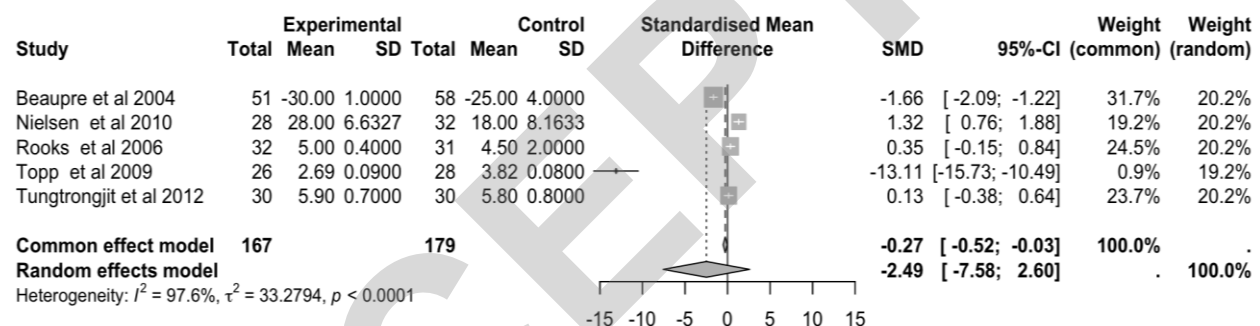


B Nutrition-Only Complication

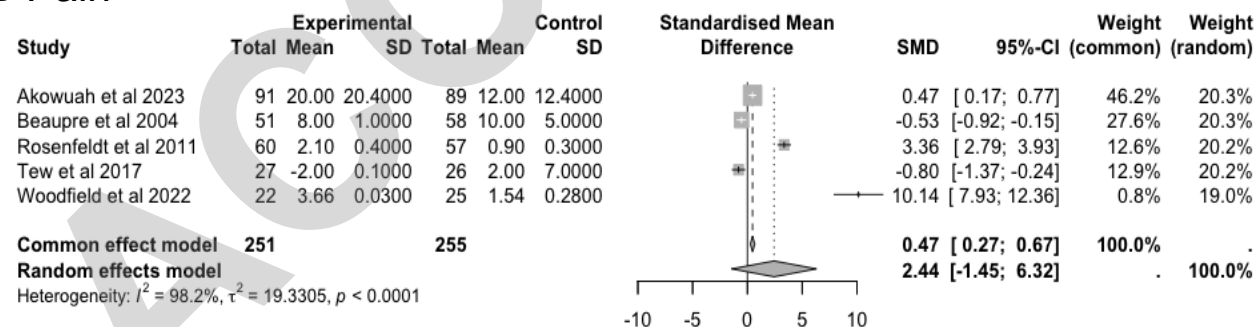
Figure 4



A Quality of Life



B Pain



C Mental Health